

The free vector two-point function in de Sitter space

Modes, covariant continuation, and the mass $M^2 = -2H^2$

Revised 10 September 2026

Abstract

The project asks for the free vector propagator at the tachyonic mass $M^2 = -2H^2$ before computing the scalar-loop correction. We first solve the positive-mass Proca problem and then examine which parts survive at the target mass. The transverse modes contain the classical constant-electric-field solution, but their Bunch–Davies mode sum is logarithmically infrared divergent. The spatial longitudinal polarization has a negative kinetic coefficient at early sub-horizon times. These facts obstruct an ordinary positive-type Bunch–Davies state for the literal local Proca theory acting on all Fourier modes. A finite de Sitter-covariant analytic bisolution nevertheless exists, and its field equation and transversality can be checked explicitly. We distinguish this object from a causal inverse and from a physical quantum state. The distinction leaves the scalar one-loop 1PI calculation intact, but requires a precise choice of the finite matching condition intended to replace the paper’s momentum-space prescription.

1 Problem and conventions

The requested object is the free vector two-point function in the project notes [1]. The reference paper [2] uses a classical dynamical vector field to sustain a constant physical electric field and explicitly does not quantize the photon. Quantizing every fluctuation of its local Proca action is therefore an additional assumption, which we make explicit throughout this note.

We use the Poincaré patch, signature $(-+++)$, and covariant components A_μ :

$$ds^2 = a^2(\eta)(-d\eta^2 + d\mathbf{x}^2), \quad a(\eta) = -\frac{1}{H\eta}, \quad \eta < 0, \quad R_{\mu\nu} = 3H^2 g_{\mu\nu}. \quad (1)$$

Here $H > 0$ is constant and $\square = \nabla^\mu \nabla_\mu$. The mass parameter M^2 is initially positive and is later set to $-2H^2$. The action is

$$S_P = \int d^4x \sqrt{-g} \left[-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} M^2 A_\mu A^\mu \right], \quad F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu. \quad (2)$$

Variation and the divergence of the field equation give

$$\nabla^\nu F_{\nu\mu} - M^2 A_\mu = 0, \quad M^2 \nabla^\mu A_\mu = 0, \quad (3)$$

$$(\square - M^2 - 3H^2) A_\mu = 0, \quad \nabla^\mu A_\mu = 0 \quad (M^2 \neq 0). \quad (4)$$

At the target mass the wave operator on the constrained field is $\square - H^2$.

A negative value of M^2 does not prevent a *real* Stueckelberg completion. Without introducing a square root of M^2 , one can write

$$\mathcal{L}_{\text{St}} = -\frac{1}{4} F^2 - \frac{M^2}{2} (A_\mu - \nabla_\mu \theta)(A^\mu - \nabla^\mu \theta), \quad \delta A_\mu = \nabla_\mu \alpha, \quad \delta \theta = \alpha. \quad (5)$$

All fields and transformations remain real when $M^2 = -\mu^2 < 0$. The problem is the sign of the scalar kinetic term:

$$-\frac{M^2}{2}(\nabla\theta)^2 = \frac{\mu^2}{2}(\nabla\theta)^2 \supset -\frac{\mu^2}{2}(\partial_0\theta)^2. \quad (6)$$

It is a wrong-sign kinetic term in a local orthonormal frame, rather than a failure to define a real gauge symmetry. Gauge fixing does not remove the physical Proca longitudinal degree of freedom.

For $M^2 > 0$, diagonalizing the usual Stueckelberg system in a covariant gauge gives the vector action

$$S_\xi[A] = \int d^4x \sqrt{-g} \left[-\frac{1}{4}F^2 - \frac{1}{2}M^2 A^2 - \frac{1}{2\xi_A}(\nabla_\mu A^\mu)^2 \right]. \quad (7)$$

Below we take $\xi_A > 0$ when quoting its Bunch–Davies two-point function. This gauge-potential correlator includes an unphysical sector; the Proca correlator will be identified separately.

2 Canonical transverse and longitudinal variables

Conformal invariance of the four-dimensional Maxwell term reduces (2) to

$$S_P = \int d\eta d^3x \left[\frac{1}{2}(A'_i - \partial_i A_0)^2 - \frac{1}{4}(\partial_i A_j - \partial_j A_i)^2 + \frac{1}{2}a^2 M^2 A_0^2 - \frac{1}{2}a^2 M^2 A_i^2 \right]. \quad (8)$$

A prime in the mode calculation denotes ∂_η . Decompose

$$A_i = A_i^T + \partial_i \chi, \quad \partial_i A_i^T = 0. \quad (9)$$

For nonzero spatial momentum $k = |\mathbf{k}|$, orthogonality gives

$$S_T = \frac{1}{2} \int d\eta \frac{d^3k}{(2\pi)^3} \left[|A_i^{T'}|^2 - (k^2 + a^2 M^2) |A_i^T|^2 \right], \quad (10)$$

$$S_L = \frac{1}{2} \int d\eta \frac{d^3k}{(2\pi)^3} \left[k^2 |\chi' - A_0|^2 + a^2 M^2 |A_0|^2 - a^2 M^2 k^2 |\chi|^2 \right]. \quad (11)$$

The two transverse polarizations need no field rescaling. The temporal variable is nondynamical and obeys

$$(k^2 + a^2 M^2) A_0 = k^2 \chi'. \quad (12)$$

Where the coefficient on the left is nonzero, substituting its solution gives

$$S_L = \frac{1}{2} \int d\eta \frac{d^3k}{(2\pi)^3} \left[K_L |\chi'|^2 - a^2 M^2 k^2 |\chi|^2 \right], \quad K_L = \frac{a^2 M^2 k^2}{k^2 + a^2 M^2}. \quad (13)$$

This agrees with the decomposition in Appendix A of the reference paper [2], specialized to a constant Maxwell kinetic coefficient. For $M^2 > 0$, $K_L > 0$ and $v_L = \sqrt{K_L} \chi$ is canonical. An integration by parts yields

$$S_L = \frac{1}{2} \int d\eta \frac{d^3k}{(2\pi)^3} \left[|v'_L|^2 - \omega_L^2 |v_L|^2 \right], \quad \omega_L^2 = k^2 + a^2 M^2 - \frac{(\sqrt{K_L})''}{\sqrt{K_L}}. \quad (14)$$

An alternative variable avoids division by $k^2 + a^2 M^2$ in the equations of motion. Define the longitudinal electric variable

$$\pi_k = \chi'_k - A_{0,k}. \quad (15)$$

The momentum conjugate to χ_k is $k^2\pi_k$. Its field equation and the temporal constraint give

$$A_{0,k} = \frac{k^2}{a^2 M^2} \pi_k, \quad \chi_k = -\frac{1}{a^2 M^2} \pi'_k, \quad (16)$$

and hence

$$\pi''_k + \frac{2}{\eta} \pi'_k + \left(k^2 + \frac{M^2}{H^2 \eta^2} \right) \pi_k = 0. \quad (17)$$

These equations remain regular at $k^2 + a^2 M^2 = 0$ for $M^2 \neq 0$ and $\eta < 0$. They are useful for distinguishing a singular reduced coordinate choice from a divergence of a constrained solution.

3 Bunch–Davies modes at positive mass

Choose transverse polarization vectors with

$$k_i \epsilon_i^{(\lambda)} = 0, \quad \epsilon_i^{(\lambda)} \epsilon_i^{(\lambda')*} = \delta_{\lambda\lambda'}, \quad \sum_{\lambda=1}^2 \epsilon_i^{(\lambda)} \epsilon_j^{(\lambda)*} = \delta_{ij} - \hat{k}_i \hat{k}_j. \quad (18)$$

With $[a_\lambda(\mathbf{k}), a_{\lambda'}^\dagger(\mathbf{q})] = (2\pi)^3 \delta_{\lambda\lambda'} \delta^3(\mathbf{k} - \mathbf{q})$, write

$$A_i^T(\eta, \mathbf{x}) = \sum_{\lambda=1}^2 \int \frac{d^3 k}{(2\pi)^3} \left[\epsilon_i^{(\lambda)} a_\lambda(\mathbf{k}) f_k(\eta) e^{i\mathbf{k}\cdot\mathbf{x}} + \text{h.c.} \right]. \quad (19)$$

The mode equation and canonical normalization are

$$f''_k + \left(k^2 + \frac{M^2}{H^2 \eta^2} \right) f_k = 0, \quad f_k f_k^{*'} - f_k^* f'_k = i. \quad (20)$$

For

$$\nu_V^2 = \frac{1}{4} - \frac{M^2}{H^2}, \quad (21)$$

the Bunch–Davies solution is

$$f_k^{\text{BD}}(\eta) = \frac{\sqrt{-\pi\eta}}{2} \exp\left[\frac{i\pi}{4}(2\nu_V + 1)\right] H_{\nu_V}^{(1)}(-k\eta), \quad f_k^{\text{BD}} \xrightarrow{-k\eta \rightarrow \infty} \frac{e^{-ik\eta}}{\sqrt{2k}}. \quad (22)$$

We use $\nu_V \geq 0$ when the order is real and $\nu_V = i\rho$, $\rho > 0$, otherwise. The exponential in (22) includes the modulus needed to normalize the imaginary-order modes.

Expand π_k with an independent longitudinal oscillator and mode g_k . For $M = \sqrt{M^2} > 0$, a convenient choice, up to a constant phase, is

$$g_k^{\text{BD}}(\eta) = \frac{M}{k} \sqrt{\frac{\pi}{4H}} (-H\eta)^{-1/2} e^{i\pi\nu_V/2} H_{\nu_V}^{(1)}(-k\eta). \quad (23)$$

The commutator of χ with its momentum $k^2\pi$ requires

$$g_k g_k^{*'} - g_k^* g'_k = \frac{ia^2 M^2}{k^2} = \frac{iM^2}{k^2 H^2 \eta^2}. \quad (24)$$

Equation (23) satisfies this identity. Together with (16), it supplies A_0 and the spatial longitudinal polarization of the positive-mass Proca mode sum. In the short-wavelength limit these three physical polarizations have the local flat-space Proca normalization.

4 The target mass: classical solution and quantum obstruction

At $M^2 = -2H^2$ the transverse order is $\nu_V = 3/2$ and

$$f_k(\eta) = \frac{1}{\sqrt{2k}} \left(1 - \frac{i}{k\eta}\right) e^{-ik\eta}. \quad (25)$$

The $k = 0$ equation and its two independent solutions are

$$A_i^{T''} - \frac{2}{\eta^2} A_i^T = 0, \quad A_i^T = \frac{c_i}{-\eta} + d_i \eta^2. \quad (26)$$

In particular, the profile used in the paper,

$$A_0 = 0, \quad A_i = \frac{E}{H^2 \eta} \delta_{iz}, \quad E_{\text{phys}} = -a^{-2} A'_z = E, \quad (27)$$

is an exact classical solution. This verifies why the mass was introduced.

At small nonzero k , $f_k(\eta) f_k^*(\eta') \sim (2k^3 \eta \eta')^{-1}$. Consequently the transverse Wightman integral would contain

$$G_{ij}^{T,+}(x, x') = \int \frac{d^3 k}{(2\pi)^3} (\delta_{ij} - \hat{k}_i \hat{k}_j) f_k(\eta) f_k^*(\eta') e^{ik \cdot (x - x')}, \quad (28)$$

$$G_{ij}^{T,+}(x, x')|_{\text{IR}} \sim \frac{\delta_{ij}}{6\pi^2 \eta \eta'} \int_{k_{\text{IR}}} \frac{dk}{k}. \quad (29)$$

The coefficient follows by angularly averaging the transverse projector. A fixed comoving infrared cutoff gives a finite integral but breaks exact de Sitter invariance. Removing one isolated Fourier mode does not remove the logarithm from the continuum of arbitrarily small momenta.

The longitudinal coefficient and temporal constraint become

$$K_L = -\frac{2k^2}{k^2 \eta^2 - 2}, \quad (k^2 - 2/\eta^2) A_0 = k^2 \chi'. \quad (30)$$

Thus $K_L < 0$ for $|k\eta| > \sqrt{2}$ and $K_L > 0$ for $|k\eta| < \sqrt{2}$. Every fixed nonzero comoving momentum begins in the negative-kinetic regime. At $|k\eta| = \sqrt{2}$, algebraic elimination of A_0 fails, but the regular system (16)–(17) still enforces the constraint. Indeed, at the target mass

$$\chi = \frac{\eta^2}{2} \pi', \quad \chi' = \left(1 - \frac{k^2 \eta^2}{2}\right) \pi, \quad (31)$$

so $\chi' = 0$ at the crossing for a regular π . A pole of the reduced coefficient is therefore not a proof that every constrained solution diverges. The early-time kinetic sign is already sufficient to expose the positive-norm Bunch–Davies problem [5].

The Wronskian makes this sign issue precise. Put $\mu = \sqrt{2}H$. Substituting $M = i\mu$ in the expression (23) and then taking *actual complex conjugates* gives

$$g_k g_k^{*'} - g_k^* g'_k = +\frac{i\mu^2}{k^2 H^2 \eta^2}, \quad \text{whereas the canonical requirement is } -\frac{i\mu^2}{k^2 H^2 \eta^2}. \quad (32)$$

A phase i cannot reverse a sesquilinear norm. Keeping the BD frequency choice therefore requires a negative oscillator commutator, hence an indefinite inner product. Choosing the conjugate frequency can instead reverse the Wronskian, but abandons the ordinary positive-frequency BD prescription for that polarization. Continuing an analytic formula in M^2 is not the same operation as continuing a positive-type mode sum.

For $k = 0$, $\partial_i \chi = 0$ and the longitudinal action vanishes. The homogeneous transverse ansatz in (27) consequently avoids this longitudinal degree of freedom. Its existence does not define a healthy quantum theory for all spatial momenta.

5 Covariant two-point function at positive mass

The positive-mass mode sum can be expressed in de Sitter-invariant form; we use the result and conventions of Fröb and Higuchi [3]. For the Poincaré patch define

$$Z_\epsilon = 1 + \frac{(\eta - \eta' - i\epsilon)^2 - |\mathbf{x} - \mathbf{x}'|^2}{2\eta\eta'}, \quad \epsilon \downarrow 0, \quad (33)$$

with its Wightman boundary value understood as in Step 1. Away from the singular support we write Z for the invariant argument. Define

$$I_\alpha(Z_\epsilon) = \Gamma\left(\frac{3}{2} + \alpha\right) \Gamma\left(\frac{3}{2} - \alpha\right) {}_2F_1\left(\frac{3}{2} + \alpha, \frac{3}{2} - \alpha; 2; \frac{1 + Z_\epsilon}{2}\right), \quad (34)$$

$$\rho_\xi^2 = \frac{9}{4} - \xi_A \frac{M^2}{H^2}. \quad (35)$$

Primes in the remainder of the covariant calculation mean derivatives with respect to Z . Introduce

$$\begin{aligned} K_{\mu\nu'}[I] &= (3I' + ZI'')(\nabla_\mu Z)(\nabla_{\nu'} Z) \\ &\quad + [(1 - Z^2)I'' - 3ZI']\nabla_\mu \nabla_{\nu'} Z. \end{aligned} \quad (36)$$

The gauge-potential and Proca Wightman functions are, respectively,

$$G_{\mu\nu'}^+(x, x'; \xi_A) = \frac{H^2}{16\pi^2 M^2} \left[K_{\mu\nu'}[I_{\nu_V}] - \nabla_\mu \nabla_{\nu'} I_{\rho_\xi} \right], \quad (37)$$

$$G_{\mu\nu'}^{P,+}(x, x') = \frac{H^2}{16\pi^2 M^2} K_{\mu\nu'}[I_{\nu_V}], \quad M^2 > 0. \quad (38)$$

The first term contains all three physical Proca polarizations and agrees with the Allen–Jacobson propagator [4]. The second is a gauge-dependent pure gradient. For the Proca function,

$$\nabla^\mu G_{\mu\nu'}^{P,+} = 0, \quad (\square_x - M^2 - 3H^2)G_{\mu\nu'}^{P,+} = 0 \quad (x \neq x'). \quad (39)$$

Equations (37) and (38) are Wightman functions, not inverse operators. A Proca Feynman inverse requires time ordering and a local contact term associated with the nondynamical temporal component [3]. A retarded inverse instead requires a causal prescription. These distinctions matter if a vector inverse is used to propagate a source, even though no internal vector inverse is needed for the scalar one-loop 1PI kernel.

6 Finite covariant continuation and its checks

To approach the target mass, put $\delta = 3/2 - \nu_V$ and $q = 1 - Z_\epsilon$. The scalar structure function has the Laurent expansion

$$I_{3/2-\delta}(Z_\epsilon) = \frac{2}{\delta} + \hat{I}_{3/2}(Z_\epsilon) + O(\delta), \quad \hat{I}_{3/2} = \frac{2}{q} - 2 \ln\left(\frac{q}{2}\right) + C. \quad (40)$$

The value of the finite constant C is irrelevant here. Every term in $K[I]$ differentiates I , so it annihilates the constant pole. In particular,

$$\hat{I}'_{3/2} = \frac{2}{q^2} + \frac{2}{q}, \quad \hat{I}''_{3/2} = \frac{4}{q^3} + \frac{2}{q^2}. \quad (41)$$

The resulting finite bitensor is

$$K_{\mu\nu'}^{(-2)} = 4(q^{-3} + q^{-2} + q^{-1})(\nabla_\mu Z)(\nabla_{\nu'} Z) + (2q^{-2} + 4)\nabla_\mu \nabla_{\nu'} Z, \quad (42)$$

$$G_{\mu\nu'}^{\text{formal}}(x, x') = -\frac{1}{32\pi^2} K_{\mu\nu'}^{(-2)}(Z_\epsilon). \quad (43)$$

This is the finite limit of the meromorphic Proca expression; it is not the infrared-regulated positive-type transverse mode sum.

The exceptional scalar finite part obeys an anomalous equation,

$$(1 - Z^2)\hat{I}_{3/2}'' - 4Z\hat{I}_{3/2}' = 6. \quad (44)$$

Indeed, the scalar equation before the limit has a mass coefficient $\delta(3 - \delta)$ multiplying $I_{3/2-\delta}$, whose pole contributes 6. It would be incorrect to treat $\hat{I}_{3/2}$ as a massless scalar homogeneous solution. The differentiated vector combination nevertheless does obey its homogeneous vector equation.

For an explicit check, write

$$T_{\mu\nu'} = P(Z)(\nabla_\mu Z)(\nabla_{\nu'} Z) + Q(Z)\nabla_\mu \nabla_{\nu'} Z. \quad (45)$$

The de Sitter identities

$$\begin{aligned} (\nabla Z)^2 &= H^2(1 - Z^2), & \square Z &= -4H^2 Z, \\ (\nabla^\mu Z)\nabla_\mu \nabla_{\nu'} Z &= -H^2 Z \nabla_{\nu'} Z, & \square \nabla_{\nu'} Z &= -4H^2 \nabla_{\nu'} Z \end{aligned} \quad (46)$$

give

$$\nabla^\mu T_{\mu\nu'} = H^2[(1 - Z^2)P' - 5ZP - ZQ' - 4Q]\nabla_{\nu'} Z. \quad (47)$$

For $P = 4(q^{-3} + q^{-2} + q^{-1})$ and $Q = 2q^{-2} + 4$, the coefficient vanishes. The two coefficients of $(\square - H^2)T$ are proportional to

$$(1 - Z^2)P'' - 8ZP' - 6P - 2Q' = 0, \quad (48)$$

$$(1 - Z^2)Q'' - 4ZQ' - 2Q - 2ZP = 0. \quad (49)$$

Both identities hold for these P, Q . Thus the formal expression is transverse and a vector bisolution away from coincidence. The same statements hold at the primed point by symmetry. This check establishes the homogeneous equations; it does not supply an inhomogeneous contact term or prove positivity.

There is also a short-distance check. At small spacelike geodesic separation ℓ , $q \sim H^2 \ell^2/2$, $\nabla Z = O(H^2 \ell)$, and $\nabla \nabla' Z = O(H^2)$. The leading terms of (43) scale as $1/(H^2 \ell^4)$ and the bitensor has mass dimension two. This is the expected strict-Proca longitudinal singularity, analogous to $M^{-2} \partial_\mu \partial_{\nu'} \Delta^+$ in flat space. The pure-gradient sector must be retained when assessing the short-distance form of the finite- ξ_A gauge potential instead.

The finite analytic expression does not repair the infrared integral (29) or the longitudinal norm in (32). For $\xi_A > 0$ the gauge scalar at the target mass has $\rho_\xi^2 = 9/4 + 2\xi_A > 9/4$, so its naive BD mode sum is infrared divergent as well. We therefore use the notation G^{formal} for (43): it is a covariant analytic bisolution, with no claim that it defines a positive-type Wightman state or a causal Green inverse.

7 What this supplies for the scalar-loop problem

The scalar contribution to the 1PI quadratic action is obtained from the second functional variation of $\Gamma_\phi^{(1)}[A]$ at $A = 0$. It depends only on scalar propagators. The full paper's treatment of A_μ as a classical field is consequently sufficient for this calculation. Its local Maxwell divergence is the quantity calculated by the heat-kernel method in Step 2. The vector sector supplies the classical free operator and external profiles on which a finite response or matching condition may be imposed.

It is useful to distinguish three statements that the word “pole” can otherwise conflate. A physical particle pole has a positive spectral interpretation and appropriate state assumptions. A formal pole of an analytically continued resolvent is an operator-theoretic statement that does not by itself establish that interpretation. A finite matching condition at a specified spectral value can also be imposed away from any pole. Failure of positive-type Proca BD quantization excludes the ordinary first interpretation for the literal model; it does not automatically exclude the other two.

In particular, normalizable individual eigenfunctions are not a general prerequisite for a subtraction scheme. Minkowski plane waves are generalized eigenfunctions, and a continuum spectral problem is normally formulated with distributions or wave packets. What has to be specified in de Sitter is the operator whose spectral parameter is used, its domain or continuation, and the boundary value of the loop kernel.

A concrete Euclidean construction illustrates this point. On the sphere S^4 of radius H^{-1} , let $Y_\mu^{(L\sigma)}$ be regular transverse vector harmonics. With δ the codifferential, the Maxwell operator on this sector is $\delta d = -\nabla_E^2 + 3H^2$, and

$$\nabla_E^\mu Y_\mu^{(L\sigma)} = 0, \quad (\delta d)Y_\mu^{(L\sigma)} = H^2(L+1)(L+2)Y_\mu^{(L\sigma)}, \quad L \geq 1. \quad (50)$$

The target Proca operator therefore has spectrum

$$(\delta d - 2H^2)Y_\mu^{(L\sigma)} = H^2L(L+3)Y_\mu^{(L\sigma)}. \quad (51)$$

There is no regular transverse zero eigenmode at an allowed integer L . De Sitter covariance diagonalizes the Euclidean scalar-loop kernel:

$$\int_{S^4} d^4x' \sqrt{g_E(x')} \Pi_{\mu\nu'}^E(x, x') Y^{(L\sigma)\nu'}(x') = \Pi_L^E Y_\mu^{(L\sigma)}(x). \quad (52)$$

Matching the quadratic coefficient at an integer $L_\star$ would define a definite Euclidean subtraction after the remaining conventions have been fixed. It is an off-shell matching condition, not a pole-residue condition. A pole prescription would require a specified continuation of the spectral data, which is additional information.

The electric profile (27) is spatially homogeneous and nonnormalizable on the infinite Poincaré patch. It cannot furnish a regular target-mass zero harmonic on S^4 , since (51) has none. This observation alone is not a universal prohibition on generalized spectral methods. It does show why there is no automatic identification of an integer Euclidean matching point with a retarded response evaluated on this external history.

Retarded and time-ordered kernels are different boundary-value objects. Their relation does not identify their convolutions with an arbitrary history; integrations by parts can retain terms on the initial hypersurface [6]. A classical response on $A_z \propto \eta^{-1}$ requires the initial time or past prescription, the scalar state, the switching or boundary data, and the relevant infrared prescription. The formal vector bisolution found above does not select those data.

There is a second, independent issue with the reference prescription. In the notation of the paper,

$$\delta_3 = -e^2 \Pi(s_0), \quad s_0 = 2H^2, \quad (53)$$

fixes a value of the polarization form factor. When $s_0 \neq 0$, a claim about the residue of a dressed propagator additionally involves the derivative of its inverse denominator. Equation (53) alone does not establish unit residue. Step 4 gives the corresponding derivative condition and distinguishes it from a finite response subtraction. This issue remains even before translating momentum-space notation into a de Sitter spectral problem.

The scope of the present result is consequently conditional and precise. If (2) with $M^2 = -2H^2$ governs every vector Fourier mode, its ordinary positive-type BD quantization fails, although the classical electric solution and the formal covariant bisolution exist. If only the homogeneous classical mode is intended, the paper has not specified a quantum theory for the remaining fluctuations. A different source mechanism, such as a time-dependent Maxwell kinetic function, need not share this Proca longitudinal action. The scalar 1PI calculation can proceed in either case. The remaining project choice is the exact finite matching problem that defines its single number δ_3 ; a retarded electric-field response is one possible prescription, not an identity forced by the free propagator calculation.

References

- [1] *The photon two-point function and the scalar-loop counterterm in de Sitter space*, project notes, Secs. 1, 3–5.
- [2] M. Bastero-Gil, P. B. Ferraz, A. Torres Manso, L. Ubaldi and R. Vega-Morales, *Classical constant electric fields and the Schwinger effect in de Sitter*, arXiv:[2508.14973v2](#).
- [3] M. B. Fröb and A. Higuchi, *Mode-sum construction of the two-point functions for the Stueckelberg vector fields in the Poincaré patch of de Sitter space*, J. Math. Phys. **55** (2014) 062301, arXiv:[1305.3421](#).
- [4] B. Allen and T. Jacobson, *Vector two-point functions in maximally symmetric spaces*, Commun. Math. Phys. **103** (1986) 669–692.
- [5] A. Golovnev, *Linear perturbations in vector inflation and stability issues*, Phys. Rev. D **81** (2010) 023514, arXiv:[0910.0173](#), Sec. 3.
- [6] K. E. Leonard, T. Prokopec and R. P. Woodard, *Covariant vacuum polarizations on de Sitter background*, Phys. Rev. D **87** (2013) 044030, arXiv:[1210.6968](#).